Existing reductions: survey and calculation

Reduction Impact Atlas | Corrected scope, 2026-11-02. This project-authored appendix replaces the earlier algorithm-goal scenario appendix.

1. Survey existing relations

The catalogue contains ten existing direct relations from newly recorded task variants, and ten existing downstream relations from frozen board targets. These are promise embeddings and applications of the polynomial-verifier-to-SAT construction. They are not twenty separate new theorems. The earlier invented efficient-algorithm routes and speculative downstream pairings are withdrawn.

A restriction embedding submits the same instance to a general problem and recovers the returned witness unchanged. A verifier encoding constructs a CNF formula for a bounded certificate and its polynomial-time verifier. A satisfying assignment supplies the original certificate; unsatisfiability preserves NO-SOLUTION. The target is unrestricted SAT, with no claim of an efficient source algorithm.

2. Known relationship versus reviewed adapter

Each record supplies a source definition, certificate bound, verifier or embedding argument, recovery rule and references. The theoretical inclusion may be known while its precise graph-model adapter still awaits independent review. Do not infer an edge merely from similar names. A binary budget does not by itself bound certificate length; the records explain caps using the explicit instance size.

3. Keep the original graph frozen

Calculations use copied graph nodes, board endpoints and documented baseline bridges. Turing and non-witness-preserving edges remain excluded. A surveyed known edge is inserted only in the temporary comparison graph. Other surveyed edges are not accumulated into the baseline. Unrelated research records therefore cannot create extra paths for the current row. Endpoint and family projection remain fixed in each comparison.

4. Two different outputs

For a direct known relation k, report coverage gain between G and G+k. For a downstream relation k: B to C and an open board parent p: A to B, first form G+k, then evaluate the parent by comparing G+k with G+k+p. Only p is hypothetical. Also report the parent impact in G and the coverage gain of k alone. The two groups are ranked separately because their score meanings differ.

5. Measures and paths

For the stated before/after graphs H0 and H1, let Rv be reachable ordered variant pairs excluding identical endpoints, and Rp their fixed projection to distinct problem families. N counts Rp(H1) minus Rp(H0). V counts Rv(H1) minus Rv(H0).

B sums new variant coverage only for family pairs already reachable in H0. For each such pair, divide its new concrete variant pairs by the fixed product of source and target family sizes. B is a sum of coverage fractions, not a probability or a research-value scale. JSON includes exact fractions.

Each group sorts by descending N, then B, then V; equal triples share a rank. D_half sums 0.5^(h-1) over new family pairs, where h is the shortest witness length. It is diagnostic only. Every witness identifies known edges separately from assumption-prefixed parent edges.

6. Attribution example

Suppose G has X to A, the literature supplies B to C, and A to B is the open candidate. Adding the known downstream relation alone creates B to C. Assuming the candidate afterward creates A to B, X to B, A to C and X to C: parent impact N=4. Without the known downstream relation its impact is N=2. The combined gain relative to G is N=5. These values are reported separately and must not be added.

7. Review and reproduction

Known does not imply new to the literature. A missing graph edge is a coverage gap, not evidence of an open mathematical problem. Generic SAT encodings can create large hub-based counts; inclusion edges can be routine. Neither is a research-priority score. The original 76-question ranking is unchanged.

Run python build.py and python -m unittest -v test_ranker.py from the published source. research_candidates.json contains relation arguments and references. research.json contains hashes, known-edge records, parent assumptions and paths. The tests check isolation, witness continuity, fractional coverage and source-package rebuilds.

Cook: polynomial computation encoding | Corrected survey page | Evidence records